

ANTHONY GRAHAM JR.

Baton Rouge, LA | (916) 698-4659 | aj.graham.2004@gmail.com

Engineering Portfolio: <https://agrahamjrportfolio.my.canva.site>

I am a Mechanical Engineering student with hands-on experience in mechanical design, CAD modeling, HVAC systems, automation environments, and mechanical disassembly and analysis. I am proficient in SolidWorks, AutoCAD, MATLAB, and technical documentation. Seeking mechanical engineering internships in automotive design & manufacturing, machine learning & robotics, and energy systems.

Technical Skills

CAD & Design: SolidWorks, AutoCAD, Fusion 360, assemblies, engineering drawings, GD&T

Mechanical Engineering: Mechanical disassembly, reverse engineering, form/fit/function analysis, mechanical systems

Automation & Controls: Conveyor systems, automated material handling, autonomous sorting systems, control wiring

Thermal & HVAC Systems: HVAC controls, facilities HVAC systems, system troubleshooting, electrical systems

Software & Analysis: MATLAB, Microsoft Excel, Microsoft Word

Engineering Methods: Design validation, technical reports, root-cause analysis, system optimization

Education

Louisiana State University (LSU)

Candidate for Bachelor of Science in Mechanical Engineering

Aug 2022 – May 2028 (Expected)

Engineering Project Experience

Reverse Engineering – Electric Tire Pump

- Disassembled and analyzed an electromechanical tire pump, evaluating mechanical components, airflow paths, and system performance.
- Created 3D CAD models and AutoCAD drawings of internal components.
- Identified design limitations and proposed mechanical design improvements.
- Documented findings in a structured engineering technical report.

Manufacturing Process Report – Casting and Fabrication in Aircraft Engines

- Researched and analyzed machining technologies utilized in the production of aircraft engines
- Highlighted aircraft engine components casting and fabrication methods are used for.
- Identified advantages and disadvantages of casting and fabrication techniques.
- Utilized models and figures to display various examples of casting and fabrication

Professional Experience

HVAC Technician — Syntergy Energy Management Solutions LLC, Baton Rouge, LA

Sep 2024 – Present

- Diagnosed, configured, and troubleshoot HVAC control systems involving sensors, actuators, electrical components, and automated building systems.
- Performed system testing and adjustments to improve operational performance.
- Support operation of heat pumps, compressors, and variable frequency drives (VFDs)
- Experience relevant to automotive thermal systems, robotics cooling, industrial automation, and cloud infrastructure thermal systems.

Fulfillment Associate (Automation Environment) — Amazon, Port Allen, LA

May 2024 – Dec 2024

- Worked in a high-volume production environment following standardized processes, quality requirements, and efficiency metrics.
- Assisted in system and mechanical assembly operations, and building automation issue resolution.
- Gained exposure to large-scale automation, robotics integration, and process optimization.

Engineering Affiliations

National Society of Black Engineers (NSBE) – LSU

- Participated in engineering workshops, industry networking, and technical development events.